# EG9x&UG96&UG95&M95 R2.0 Reference Design

**LTE Module Series**

Rev. EG9x&UG96&UG95&M95 R2.0_Reference_Design_Rev.A

Date: 2018-02-09

Status: Released

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# About the Document

## History

| Revision | Date | Author | Description |
|---|---|---|---|
| A | 2018-02-09 | Felix YIN | Initial |

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# 1 Reference Design

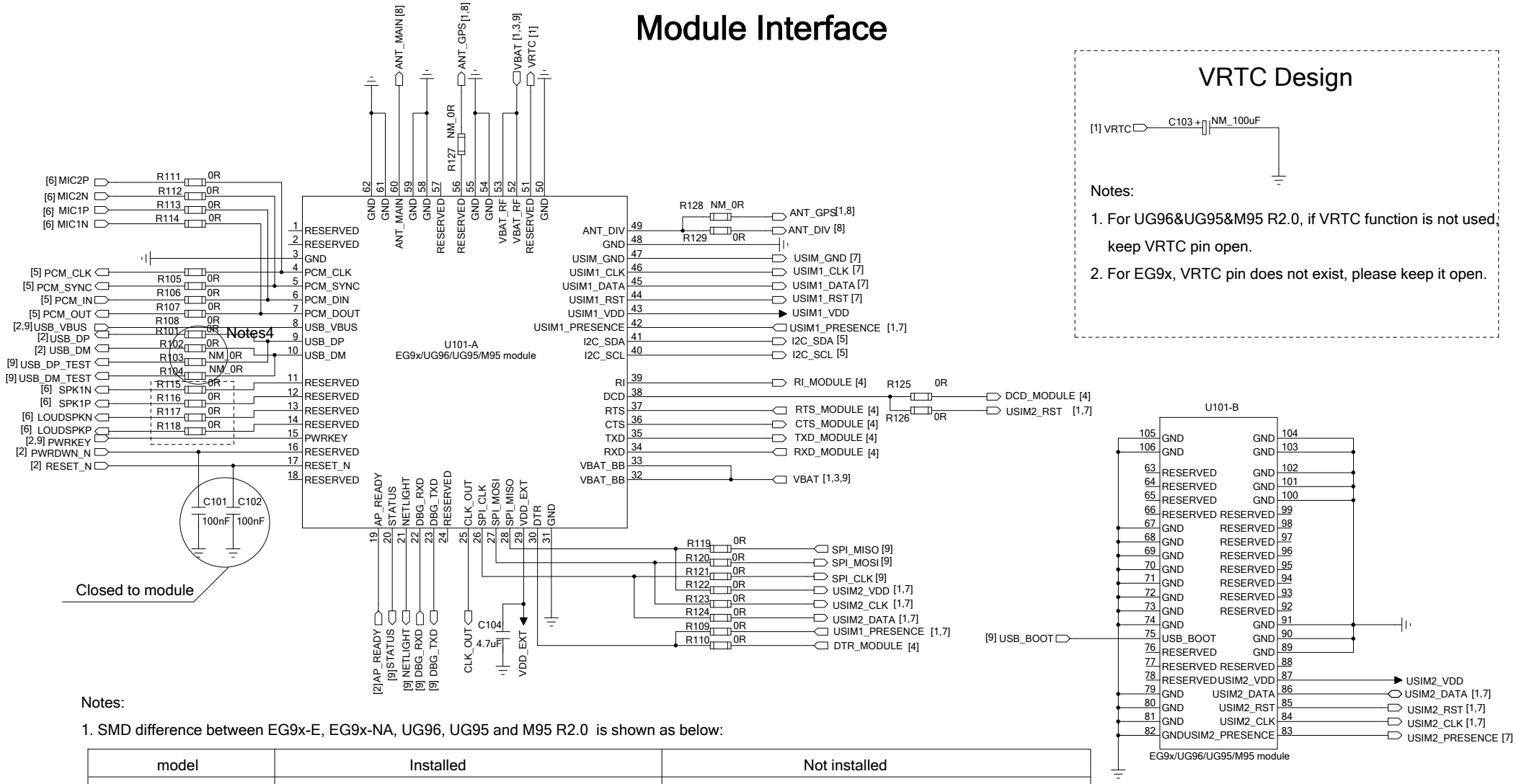
## 1.1. Introduction

This document provides the reference design for Quectel EG9x&UG96&UG95&M95 R2.0 modules, in which EG9x contains EG91 and EG95 modules.

## 1.2. Schematics

The schematics illustrated in the following pages are provided for your reference only.

# Module Interface




## VRTC Design




Notes:

1. For UG96&UG95&M95 R2.0, if VRTC function is not used, keep VRTC pin open.
2. For EG9x, VRTC pin does not exist, please keep it open.

Notes:

1. SMD difference between EG9x-E, EG9x-NA, UG96, UG95 and M95 R2.0 is shown as below:

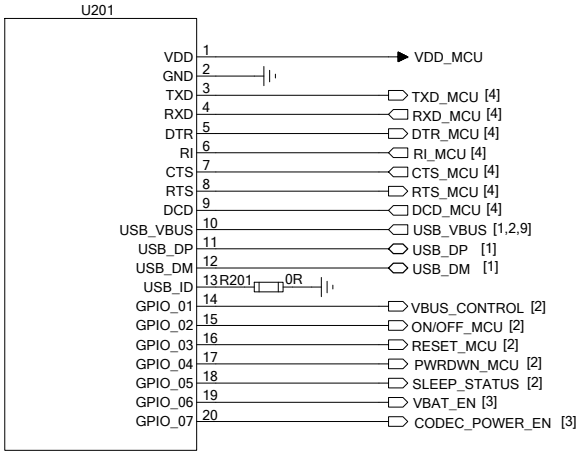
| model | Installed | Not installed |
|---|---|---|
| EG9x-E | R105～R108, R119～R121, R110, R125, R129 | C101, R111～R118, R109, R122～R124, R126～R128 |
| EG9x-NA | R105～R108, R119～R121, R110, R125, R127～R128 | C101, R111～R118, R109, R122～R124, R126, R129 |
| UG96/UG95 | C101, R105～R108, R110, R125 | R111～R118, R119～R124, R109, R126～R129 |
| M95 R2.0 | R111～R118, R109, R122～R124, R126 | R105～R108, R119～R121, R110, R125, C101, R127～R129 |

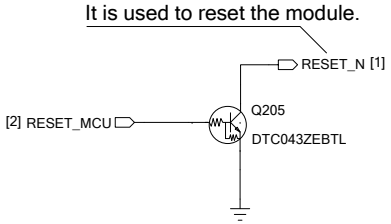
2. Please refer to EG91, EG95, UG96, UG95 and M95 R2.0 Hardware Designs respectively for further details about the differences between them.
3. Keep all RESERVED and unused pins unconnected.
4. It is recommended to reserve the test points for upgrading the firmware over USB interface and minimizing stub length of USB test signals.
5. EG9x/UG96/UG95 module provides a digital clock output (CLK_OUT) for an external audio codec, please refer to EG91, EG95, UG96 and UG95 Hardware Designs for further details.

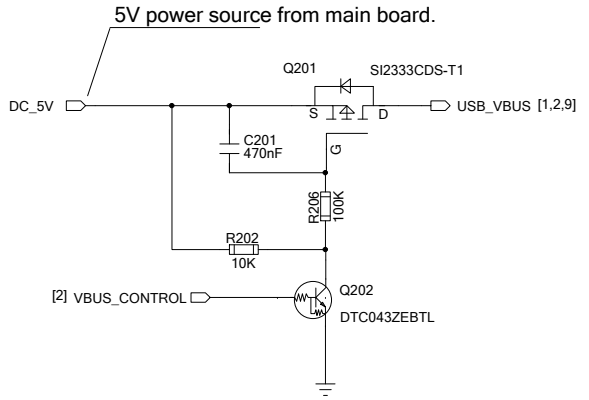
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# MCU Interface




For EG9x&M95 R2.0,

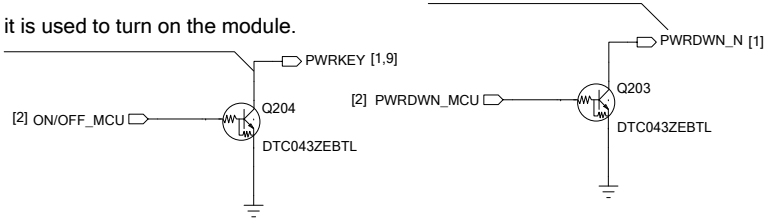
it is used to turn on or off the module.

For UG96&UG95,

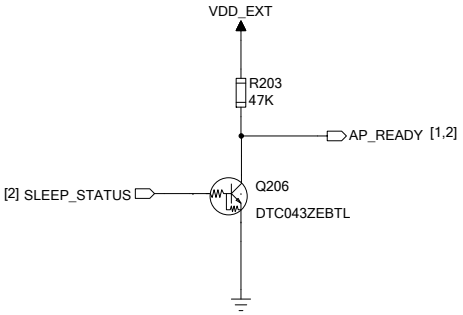
it is used to turn on the module.

For UG96&UG95,

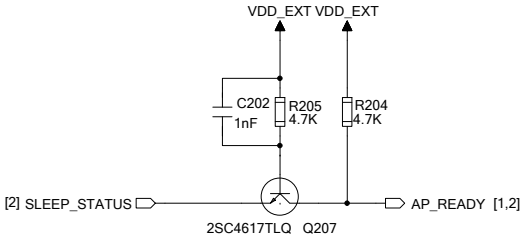
it is used to turn off the module.




The Necessary Control Circuit




AP_READY High Level Detection




AP_READY Low Level Detection

Notes:

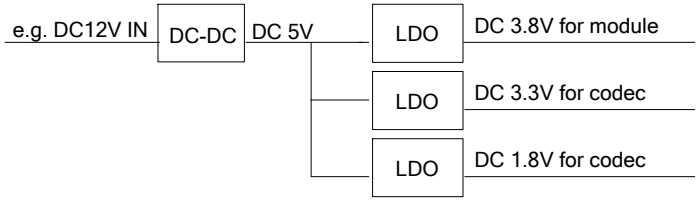
1. U201 represents customer's MCU.
2. EG9x&UG96&UG95 can only work as a USB device and supports FS/HS modes. To communicate with USB interface, MCU needs to support USB host or OTG function.
   The VBUS pins of MCU and EG9x&UG96&UG95 should be provided by 5V power system for USB detection, and VBUS_CONTROL is usded to turn on and off VBUS power supply.
   VBUS_CONTROL is active high.
3. AP_READY is used to detect the MCU's sleep state. For more details about AP_READY, please refer to EG91, EG95, UG96 and UG95 Hardware Designs and AT Commands Manuals.
4. Transistor circuits (Q203～Q207) are used for level translation.

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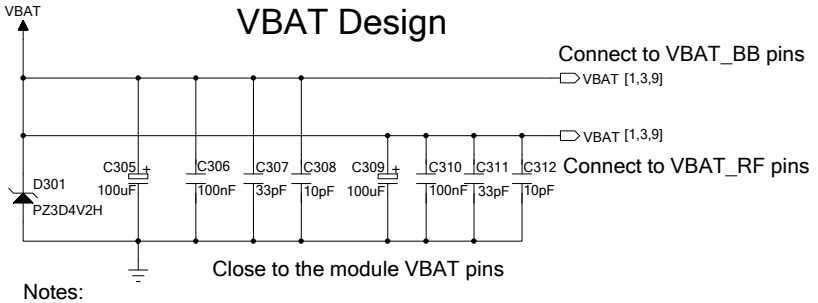
# Power Supply Design

## DC-DC Application

It is used when the input voltage is above 7V. Use a DC-DC convertor to convert a high input voltage to a 5V output, and then the LDOs will generate a 3.8V typical voltage for the module and 1.8V&3.3V typical voltages for codec.




## VBAT Design

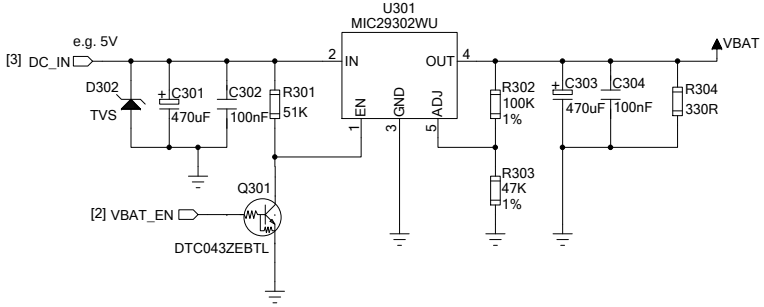



Notes:

1. The power supply must be able to provide sufficient current up to 2A or more.
2. VBAT should be routed in star mode to VBAT_BB and VBAT_RF pins.
3. The recommended operating voltage of VBAT is 3.3V～4.3V.

## LDO Application

It is used when the input voltage is below 7V.

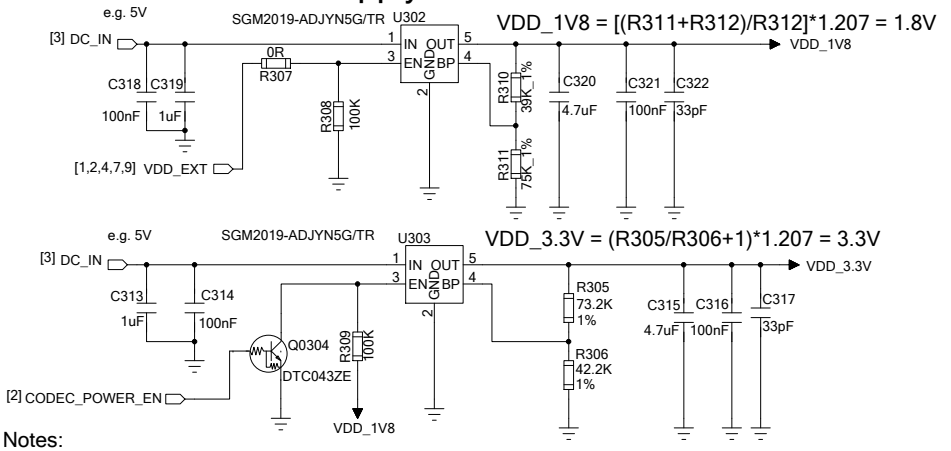



VBAT = (R302/R303+1)*1.24 = 3.88V

Notes:

1. The load current is recommended to be greater than 10mA.
2. The power supply must be able to provide sufficient current up to 2A or more.

## Supply Power for PCM Codec



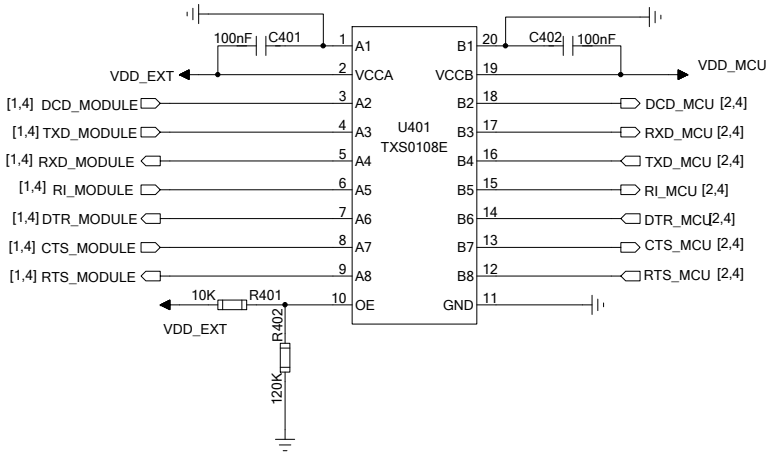

Notes:

1. CODEC_POWER_EN must be at low level in order to ensure the normal output voltage of VDD_3.3V.
   If VDD_3.3V power supply needs to be switched off, please keep CODEC_POWER_EN at high level.
2. The following power-on/off sequences should be complied with to ensure the audio codec works normally.
   Power-on Sequence: power on VDD_1V8 first, then VDD_3.3V.
   Power-off Sequence: power off VDD_3.3V first, then VDD_1V8.

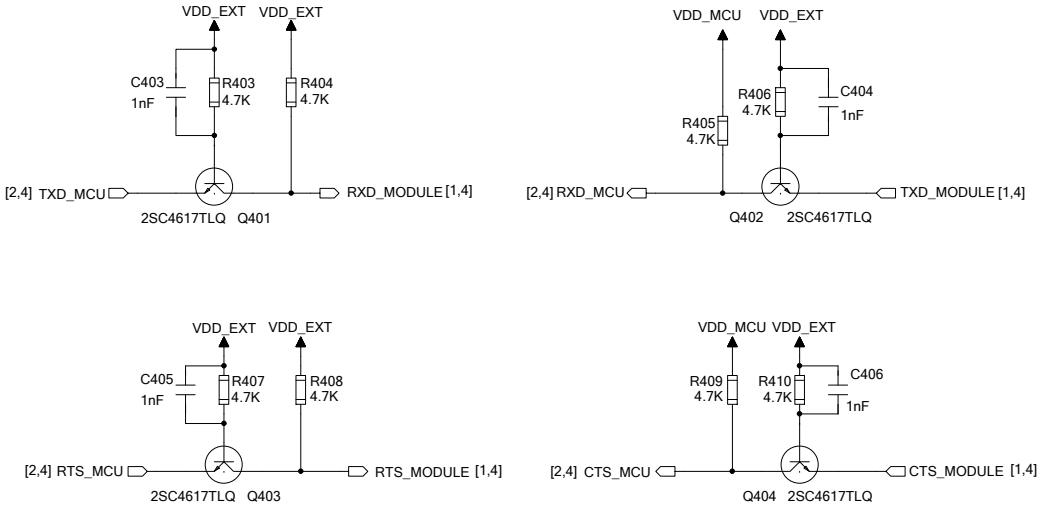
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# UART Design

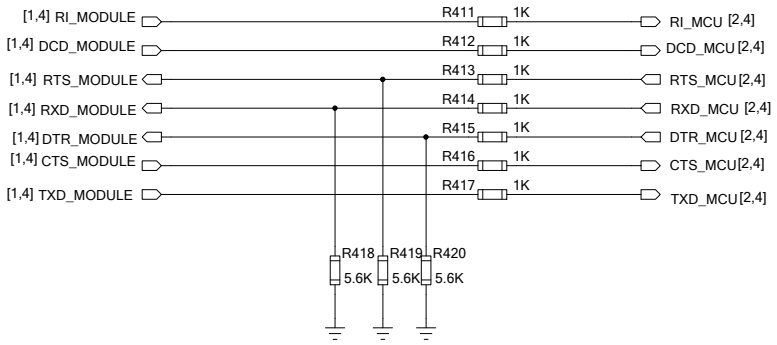
## UART Design-Option 1




## UART Design-Option 3




## UART Design-Option 2

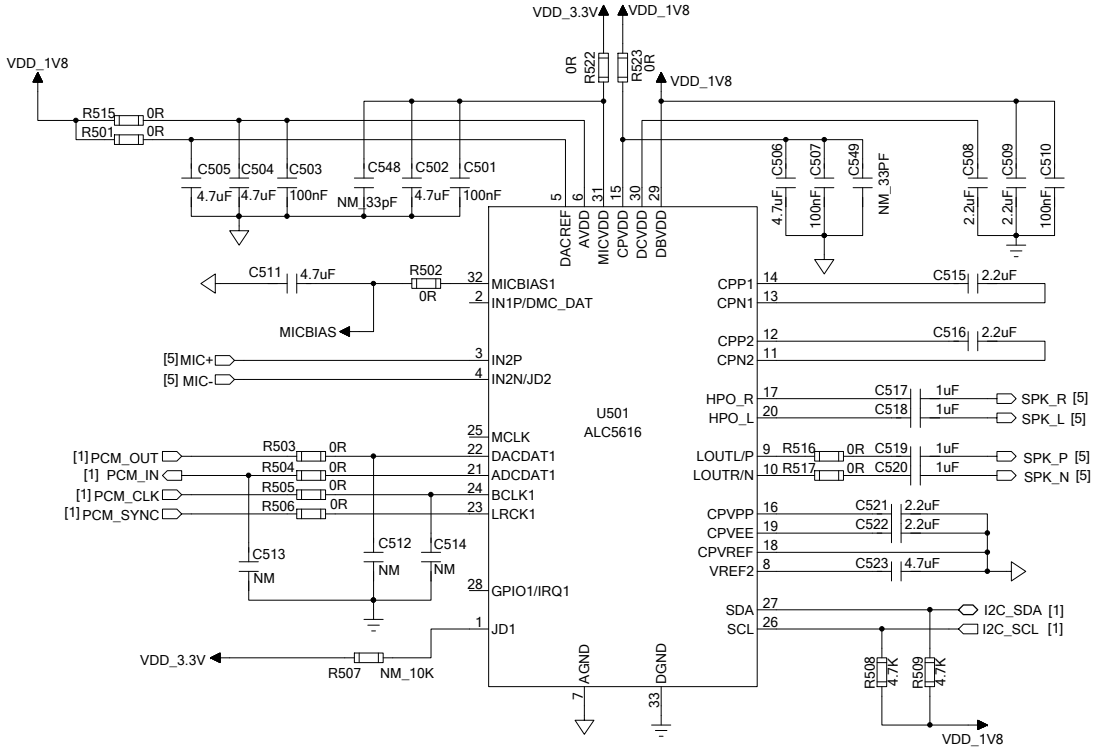


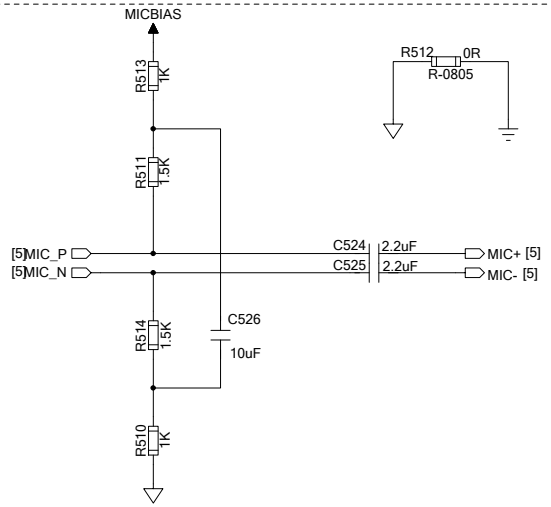

Notes:

1. The power supply voltage of VCCA should not exceed that of VCCB. For more information about TXS0108E, please refer to the datasheet from TI.
2. If a high baud rate is needed, it is highly recommended to install a 1nF capacitor (C403/C404) on transistor circiut.
3. The transistor circuit of DTR is similar to that of RTS interface, and the transistor circuit of RI and DCD is similar to that of CTS interface.
4. When EG9x/UG96/UG95 only uses USB port to communicate and M95 R2.0 uses UART port, it is recommended to use the UART design-option 2 for M95 R2.0.
5. For M95 R2.0, the resistance of R418～R420 is used on 3.3V IO level system. While on 3V IO level system, the resistance of R418～R420 should change to 15K ohm, and make sure the max input level voltage to module is 2.8V.
6. If EG9x/UG96/UG95 and M95 R2.0 module use UART port, it is recommended to use UART design-option 1 and UART design-option 3 for UART compatible design.

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# PCM Design

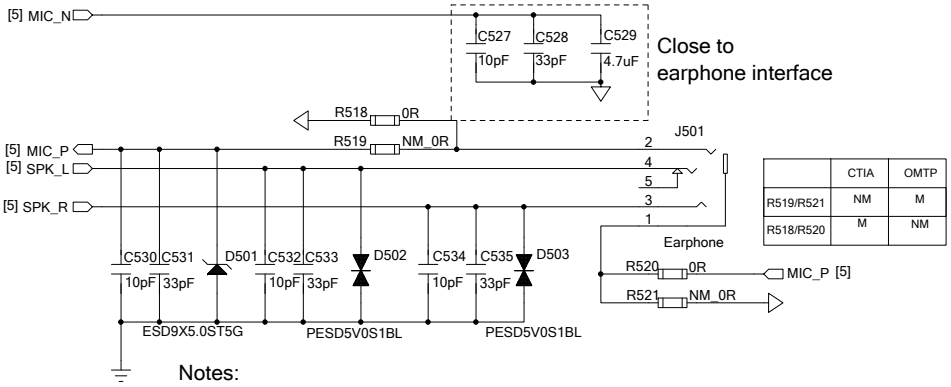



Notes:

1. ALC5616 power-on sequence: DBVDD/I2C pull-up power/AVDD /DACREF/CPVDD -> MICVDD -> software initialization.
2. ALC5616 power-off sequence: close codec function by software -> MICVDD -> DBVDD/I2C pull-up power/AVDD/DACREF/CPVDD.
3. EG9x&UG96&UG95 module will automatically initialize the codec via I2C interface after it is turned on successfully, so all power supplies for the codec need to be powered on before that.

## Audio - Earphone Application

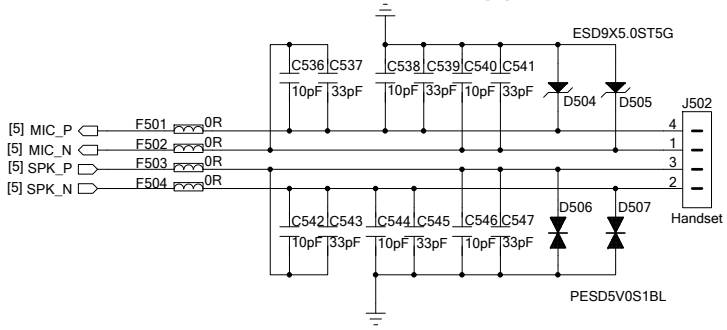



| | CTIA | OMTP |
|---|---|---|
| R519/R521 | NM | M |
| R518/R520 | M | NM |

Notes:

1. An audio amplifier should be added when the speaker impedance is less than 32 ohm.
2. The maximum capacitive loading for SPK is 330pF and the maximum capacitive loading for MIC is 250pF.

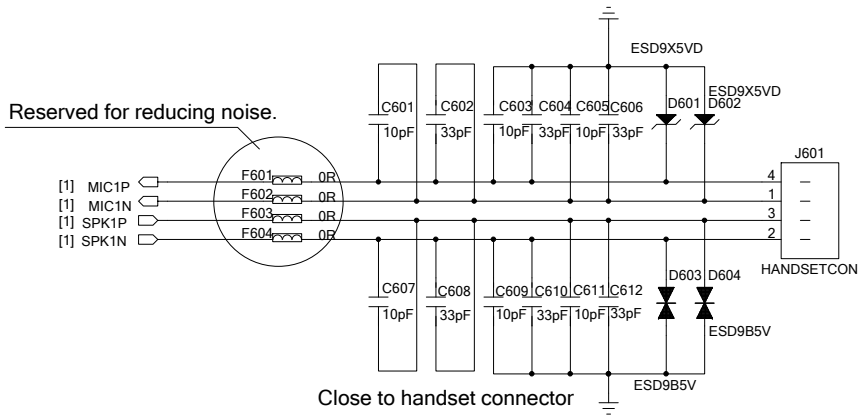
## Audio - Handset Application



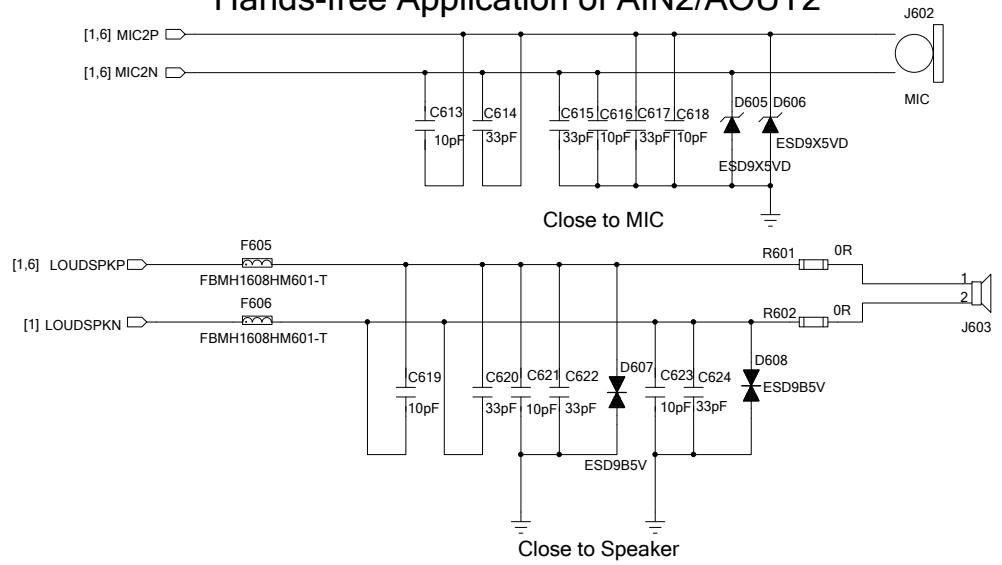

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# Analog Audio Design

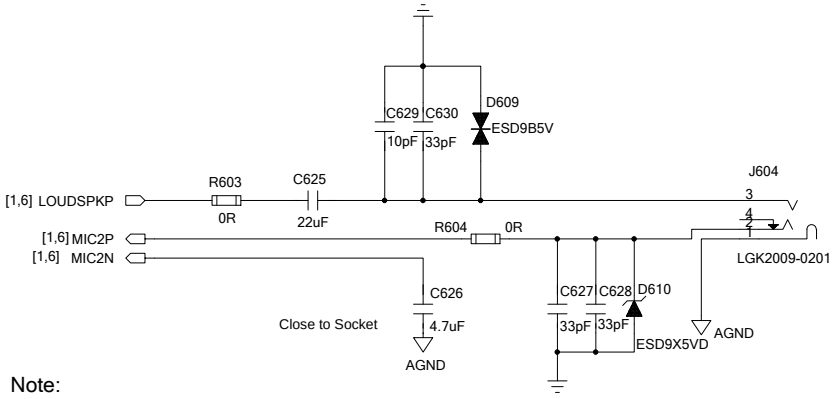
## Handset Appliocation of AIN1/AOUT1




## Hands-free Application of AIN2/AOUT2




## Earphone Application of AIN2/AOUT2




Note:

Ring function for incoming calls is only supported by AOUT2 channel.

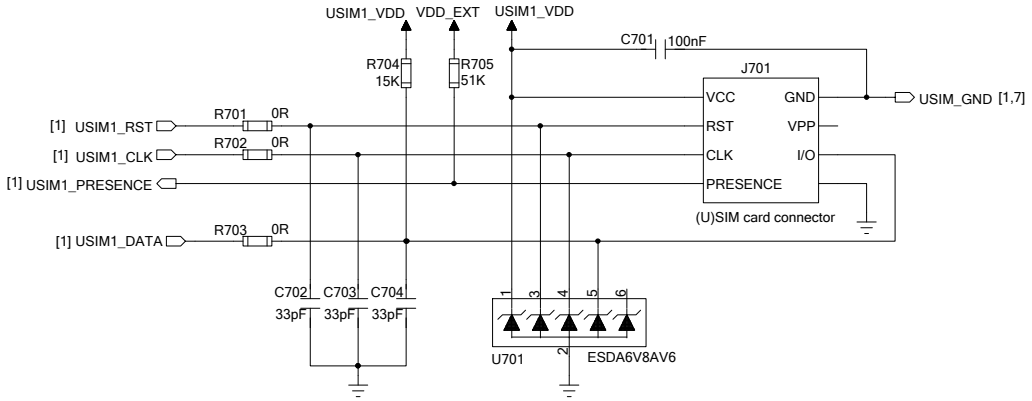
Notes:

1. Analog audio interface is for M95 R2.0 only.
2. This is the typical audio application. For more details, please refer to the *Quectel_GSM_Module_Audio_Design_Guide.*
3. 10pF&33pF capacitors are used for filtering TDD noise.
4. Both AIN1 and AIN2 have microphone bias voltage inside the module.
5. AOUT1 is capable of driving 32 ohm load.
6. AOUT2 is capable of driving 8 ohm load and earphone.

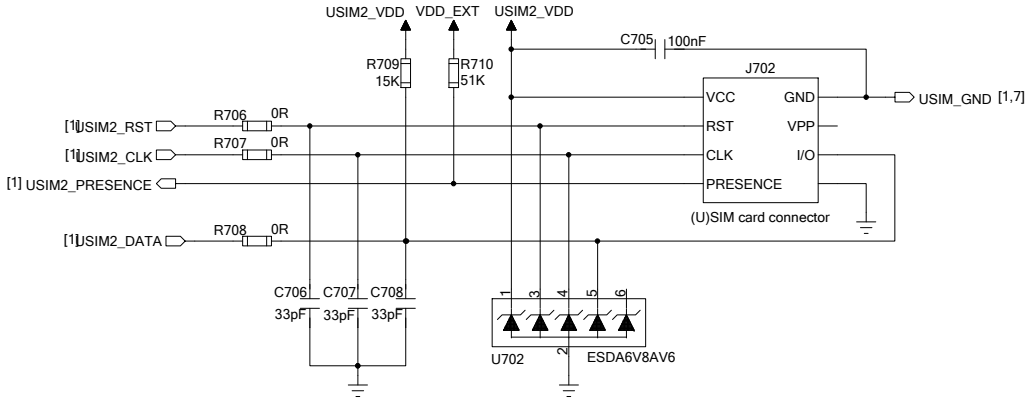
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# (U)SIM Design

## USIM1 Interface




## USIM2 Interface




Notes:

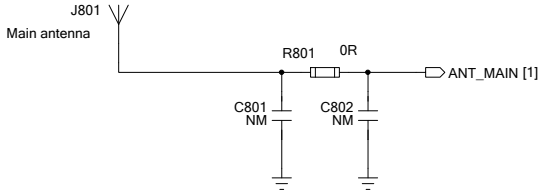
1. The decouple capacitor of USIM1_VDD and USIM2_VDD should be less than 1uF and must be near to (U)SIM card connector.
2. EG9x&UG96&UG95&M95 R2.0 module provides an input pin (USIM1_PRESENCE) to detect whether the USIM1 card is present or not. EG9x&UG96&UG95 supports both low level and high level detections, while M95 R2.0 only supports low level detection.
3. USIM2_PRESENCE input pin of EG9x module can also be used to dectect whether the USIM2 card is present or not. It supports both low level and high level detections.
4. R701～R703 and R706～R708 are used for debugging, C702～C704 and C706～C708 are used for filtering interference of GSM900MHz.
5. It is recommended to connect the (U)SIM card connector GND to the module USIM_GND. If the ground is complete on customers' PCB, USIM_GND can be connected to PCB ground directly.
6. U701 and U702 is recommended to be used to offer good ESD protection, and the parasitic capacitance should not be more than 15pF.
7. R704 and R709 can improve anti-jamming capability of the (U)SIM circuit, and it should be placed close to the (U)SIM card connector.
8. (U)SIM interface difference between EG9x, UG96, UG95 and M95 R2.0 is shown as below.

| Model | Supported (U)SIM Interface |
|---|---|
| UG96/UG95 | USIM1 |
| EG9x/M95 R2.0 | USIM1 and USIM2 |

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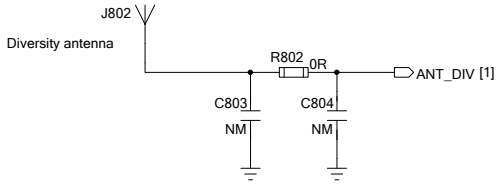
# RF and GNSS Designs

## Main Antenna Interface



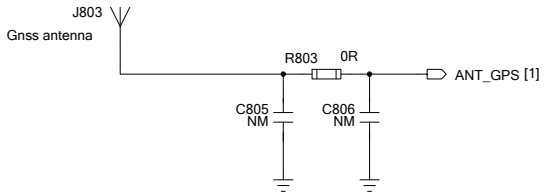

C801 and C802 are reserved for impedance matching.

## Diversity Antenna Interface




C803 and C804 are reserved for impedance matching.

## Gnss Antenna Interface

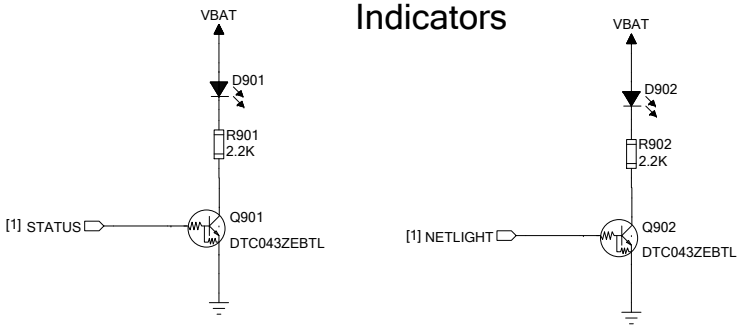



C805 and C806 are reserved for impedance matching.

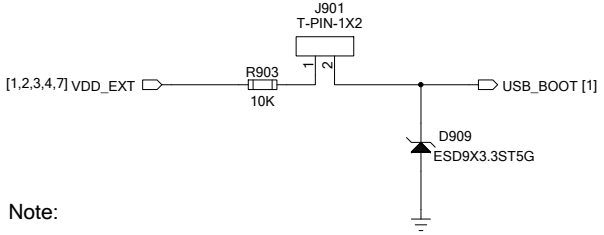
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# Indicator and Test Points

## Indicators



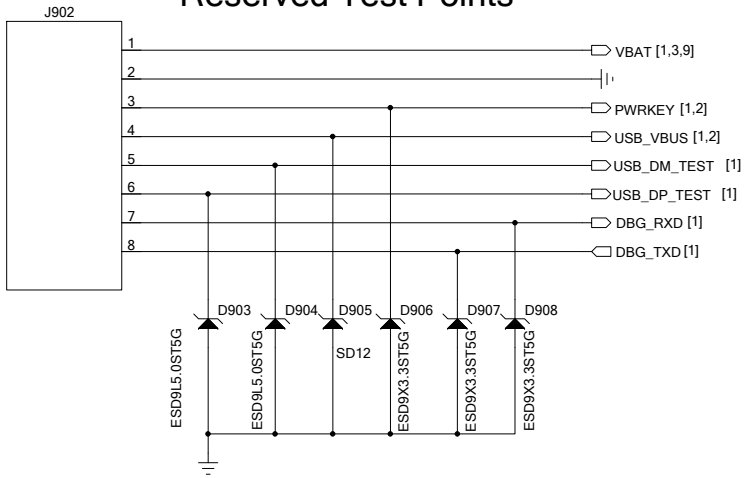

## Emergency Download Mode




Note:

When USB_BOOT is at high level, the module will be forced to enter into download mode.

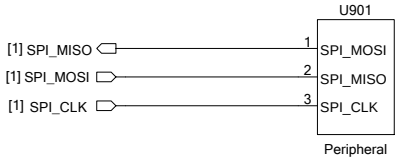
## Reserved Test Points




Notes:

1. Both USB and debug UART interface are reserved for software debugging.
2. USB interface can also be used to upgrade firmware.
3. Keep USB test points to USB pins as close as possible.
4. Please note that junction capacitance of ESD protection components on USB data lines might influence the signal. Typically, the capacitance should be less than 1pF.
5. UG96/UG95 does not have DBG interface, please keep these pins open.
6. EG9x/M95 R2.0 has DBG interface, please pay attention to level match when connecting.

## SPI Connection




Notes:

1. It is a dedicated one-to-one connection and no chip select.
2. UG96&UG95&M95 R2.0 do not have SPI interface, please keep these pins open.

| Quectel Wireless Solutions | | |
|---|---|---|
| DRAWN BY Felix YIN | PROJECT EG9x&UG96&UG95&M95 R2.0 | TITLE Reference Design |
| CHECKED BY Yeoman CHEN | SIZE A2 | VER A |
| | SHEET 9 OF 9 | DATE 2018-02-09 |